

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Bachelor of Technology / Computer Applications Major Project

A PROJECT REPORT ON **EVENT PLANNING SYSTEM** (EventHub — Real-Time Campus Event Management Platform)



Submitted in partial fulfillment of the requirements for the degree of
Bachelor of Technology in Computer Science & Engineering

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CERTIFICATE OF APPROVAL

This is to certify that the project entitled "**EVENT PLANNING & MANAGEMENT SYSTEM (EVENTHUB)**" submitted by **Priyanshu Kumar Maurya** (Roll No: _____) in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science & Engineering** is an authentic record of bonafide project work carried out under my supervision and guidance.

To the best of my knowledge, the matter embodied in this project report has not been submitted to any other University or Institute for the award of any degree or diploma.

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CANDIDATE'S DECLARATION

I hereby declare that the work presented in this project report entitled "**EVENT PLANNING SYSTEM (EVENTHUB)**" is my own original work conducted under the supervision of my faculty guide. I have complied with all academic integrity standards, cited all external resources and libraries appropriately, and not indulged in any plagiarism.

Priyanshu Kumar Maurya

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ACKNOWLEDGEMENT

I express my deep sense of gratitude and sincere respect to my project guide for their invaluable guidance, constant motivation, and constructive critique throughout the planning and execution of this web application.

I am also immensely grateful to our Head of Department and all faculty members of the Computer Science and Engineering Department for providing modern laboratory infrastructure, development environments, and technical assistance. Lastly, I extend my heartfelt thanks to my parents and peers who continually encouraged and supported me.

ABSTRACT

In contemporary academic campuses, organizing, publicizing, and participating in college events — ranging from technical hackathons, cultural festivals, sports tournaments to entrepreneurial summits — is frequently hampered by fragmented communication channels, manual paper registrations, lack of real-time seat tracking, and absence of unified attendee management for organizers.

The **Event Planning System (EventHub)** is an enterprise-grade full-stack web application architected using the **MERN (MongoDB, Express.js, React.js, Node.js)** stack to streamline college event management into an automated, real-time, paperless ecosystem. The system features:

- **Role-Based Authentication:** Secure user authentication using bcrypt password hashing and JSON Web Tokens (JWT), segregating standard student users from college administrative authorities.
- **Live MongoDB Event Inventory:** Real-time event discovery with debounced keyword search, category filtering, and sorting capabilities.
- **Strict Registration Guard:** Intelligent prevention of unauthenticated participation and duplicate registrations, backed by automatic seat quota decrement and waitlist computation.
- **Event Creation & Live Preview Studio:** Comprehensive event publication form with live card preview, future-date enforcement (preventing past date scheduling), and universal media support (including automated Instagram post conversion, Google Drive sharing, and CDN images).
- **Dynamic Event Modification:** In-place event timing, venue, capacity, and detail editing accessible to creators and administrators.
- **Administrative Intelligence & CSV Export:** Real-time platform analytics, student directory, event inventory control, and 1-click attendee roster CSV generation for campus administration.
- **Ultra-Responsive Glassmorphism Interface:** Dark purple theme optimized for mobile smartphones, tablets, and desktop displays with touch-friendly controls.

The application has been deployed on cloud infrastructure (Vercel serverless environment with MongoDB Atlas), offering instantaneous accessibility to students and college coordinators.

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CHAPTER 1: INTRODUCTION

1.1 Background

Higher education institutions thrive on extracurricular and co-curricular vitality. Throughout any academic semester, colleges and universities host a wide spectrum of events including technical symposiums, hackathons, robotics competitions, cultural evenings, debating tournaments, inter-college athletic meets, and enterprise summits. These events play a pivotal role in nurturing leadership, technological aptitude, cultural appreciation, and interpersonal collaboration among students.

However, the administrative overhead required to plan, promote, register, and coordinate such events has conventionally relied upon obsolete manual modalities. Event flyers circulated on notice boards or scattered messaging groups fail to provide comprehensive agendas, transparent seat limits, or verified attendee rosters. As student populations in contemporary educational establishments expand into thousands, the absence of an integrated, automated software framework leads to systemic chaos, registration collisions, poor student turnout, and unaccountable administration.

1.2 Problem Statement

The core problem addressed in this project is the absence of a unified, real-time digital platform tailored for college campuses that enables students to discover upcoming activities, allows society coordinators to publish and modify events with live previews, prevents unauthorized registrations, and equips administrative personnel with instant analytic reporting and attendee auditing.

1.3 Proposed System (EventHub)

To overcome these systemic shortcomings, the **Event Planning System (EventHub)** is developed as a modern, reactive single-page web application (SPA) integrated with a high-throughput NoSQL document database. EventHub provides students with an intuitive portal to explore events categorized by domain (Technology, Cultural, Sports, Business, Art, Music, Food), register with automated credential binding, monitor personal registrations, and create student-led club events.

Simultaneously, administrators are granted privileged oversight into platform metrics, attendee contact records, and automated CSV export capabilities, establishing an end-to-end, paperless operational pipeline.

CHAPTER 2: LIMITATION OF EXISTING SYSTEM

Prior to the development of the automated Event Planning System, universities and colleges predominantly leveraged conventional approaches such as Google Forms, paper registration slips, WhatsApp broadcast messages, and static notice boards. A critical analysis of these mechanisms reveals numerous operational and security limitations:

1. **Lack of Real-Time Seat Synchronization:** Conventional Google Forms or manual registers cannot dynamically enforce maximum seating limits. If an auditorium accommodates 200 participants, forms frequently continue accepting entries beyond capacity, leading to severe overbooking, seat disputes, and embarrassment for the organizing committee.
2. **Zero Authentication & Spam Entries:** In existing Google Form links shared across social media, any person can submit arbitrary, fake, or duplicate entries using anonymous email addresses without verification. There is no cryptographic authentication verifying whether the registrant is a genuine enrolled student.
3. **No Unified Schedule & Central Repository:** Information regarding different college societies (e.g., Coding Club, Dance Society, Sports Council) is siloed across disparate Instagram pages and WhatsApp channels. Students frequently miss registration deadlines due to lack of a centralized campus calendar.
4. **Absence of Dynamic Event Modification:** If an event's schedule, timing, or venue changes (e.g., rain forcing an outdoor event into an indoor seminar hall), existing posters and shared messages cannot be updated dynamically. Students remain unaware of rescheduling, resulting in disorientation.
5. **Manual Attendee Verification & Audit Challenges:** Faculty coordinators must manually stitch together multiple spreadsheets to prepare attendee lists. Exporting verified rosters with phone numbers, student college names, and timestamps is arduous and prone to human error.
6. **Poor Mobile User Experience:** Traditional forms and college portals lack responsive glassmorphism interfaces, forcing students on smartphones to pinch, zoom, and navigate clunky desktop views.

CHAPTER 3: OBJECTIVES

The primary objective of developing the **Event Planning System (EventHub)** is to engineer a dependable, scalable, and responsive web application that digitizes campus event management. The specific objectives of the project are:

- **Automated Real-Time Registration:** Implement an instant 1-click registration mechanism that records attendee profile details (name, email, phone, college) into MongoDB and automatically increments participant counts while updating seat availability indicators.
- **Strict Authentication & Role-Based Access Control (RBAC):** Establish a secure security boundary utilizing JSON Web Tokens (JWT) and bcrypt password encryption, ensuring unauthenticated visitors cannot register for events and distinguishing Student users from College Administrators.
- **Dynamic Event Creation & Live Card Preview Studio:** Provide authorized student organizers with a structured multi-section event creation studio featuring real-time previewing, tag assignment, price configuration, and custom media binding.
- **Validation Guard against Past Date Scheduling:** Enforce calendar validation constraints prohibiting organizers from scheduling events with past dates, ensuring only current and future events are permitted.
- **Universal Media & Social Link Normalization:** Implement automatic link transformation supporting Instagram post URLs (extracting high-resolution media), Google Drive public shares, and CDN images with no-referrer policies and automatic category fallbacks.
- **Event Schedule & Venue Modification:** Deliver full lifecycle event editing allowing organizers and administrators to adjust event dates, start/end times, venue rooms, and descriptions with automatic database synchronization.
- **Administrative Roster Management & 1-Click CSV Export:** Provide college administration with real-time statistics (total hosted events, aggregate registrations, registered students, estimated revenue) alongside instantaneous CSV file generation for offline verification and certificate generation.

CHAPTER 4: SCOPE

4.1 Functional Scope

The functional scope encompasses all operational modules, interactions, and business rules executed by the system:

- **User Management Module:** Registration, login, profile identification, encrypted token storage in browser local storage, and automated session renewal.
- **Event Catalog & Exploration Module:** Live search debounced at 300ms, multi-category pill filter (All, Technology, Cultural, Sports, Business, Art, Music, Food), and multi-criteria sorting (Date, Popularity, Free First, Newest).
- **Registration & Capacity Enforcement Module:** Enforces login before registration; displays real-time percentage capacity fill bars; blocks registration when seats reach capacity; prevents duplicate participation.
- **Event Creation & Modification Module:** Input fields for Title, Category, Date, Time, Venue, Seats, Price, Image URL, Tags; live sticky card preview; date picker locked to `min=Today`; in-place edit capability for event creators and admins.
- **Student Dashboard Module:** Segregated tabs displaying "Events You Registered For" with 1-click unregister, and "Events You Organized" with quick edit and delete actions.
- **Admin Dashboard Module:** Comprehensive metrics dashboard, searchable event inventory, per-event attendee directories, registered user roster, and automated browser-side CSV export generator.

4.2 Technical Scope

The technical architecture adheres to contemporary web engineering practices:

- **Client-Side Architecture:** Developed using React 18, utilizing functional components, custom hooks, and context API (`AuthContext`) for global state management without external heavy libraries.
- **Build Optimization:** Powered by Vite build tool providing sub-second Hot Module Replacement (HMR) and roll-up production minification.

- **Server-Side Architecture:** RESTful API engine powered by Node.js and Express.js, featuring modular routing (`authRoutes`, `eventRoutes`) and centralized error handling middleware.
- **Database Architecture:** Hosted on MongoDB NoSQL database utilizing Mongoose Object Data Modeling (ODM) with embedded sub-documents for registered attendees.
- **Security Infrastructure:** JWT bearer tokens, bcrypt hashing (10 salt rounds), CORS whitelisting, and input sanitization.

CHAPTER 5: TECHNOLOGY STACK DETAILED ANALYSIS

The selection of technologies for the Event Planning System was governed by criteria of performance, developer ergonomics, security, and cloud scalability. The system utilizes the MERN stack:

Layer	Technology	Role in EventHub
Frontend Library	React.js (v18.3)	Declarative component-based UI rendering, virtual DOM reconciliation, stateful UI hooks (useState, useEffect, useMemo).
Build Tool & Bundler	Vite (v8.2)	Next-generation frontend tooling offering rapid development server startup, ES modules resolution, and optimized production bundle.
Routing Engine	React Router DOM (v6.22)	Client-side single-page routing, URL parameter extraction (/events/:id, /events/:id/edit), and protected route redirects.
Iconography & UI	React Icons (Feather Icons)	Lightweight, scalable vector icons (FiCalendar, FiClock, FiMapPin, FiUsers, FiLock, FiEdit2).
Styling & CSS	Vanilla CSS3 (Glassmorphism)	Custom design system utilizing CSS Custom Properties (Variables), backdrop-filters, CSS Grid, Flexbox, and fluid typography clamp().
Backend Runtime	Node.js (v22.x LTS)	Asynchronous, non-blocking I/O runtime executing server-side JavaScript and handling concurrent network requests.
Backend Framework	Express.js (v4.19)	Minimalist REST API routing engine, middleware chaining (CORS, body-parser, auth guards), and HTTP error routing.

Layer	Technology	Role in EventHub
Database Engine	MongoDB (NoSQL)	High-performance document database storing BSON records for users and events with embedded attendee schemas.
Data Modeling	Mongoose (v8.3)	Schema validation, relationship modeling, pre-save cryptographic hooks, and query projection.
Security & Tokens	JWT & bcryptjs	Stateless authentication tokens passed via HTTP Authorization headers; secure salt-hashed password storage.

CHAPTER 6: ADVANTAGES

Implementing the Event Planning System offers substantial operational, administrative, and ecological benefits over legacy methodologies:

1. **Zero Overbooking with Automated Seat Locks:** The platform dynamically monitors attendee counts against declared capacity. Once spots are exhausted, the registration button automatically transitions to a disabled "Event Full" state, preventing seat collisions.
2. **100% Paperless and Eco-Friendly:** Eliminates physical registration desks, paper forms, and printed ticket stubs. All records, registrations, and verification lists reside securely in digital storage.
3. **Instant Administrative Reporting (CSV Generation):** Campus authorities can download comprehensive attendee lists containing student names, contact numbers, email addresses, and timestamps with a single click, saving hours of manual consolidation.
4. **Universal Social Media Image Integration:** Organizers can directly paste Instagram post links, Google Drive photo links, or Unsplash URLs without needing to re-upload files to expensive third-party storage buckets.
5. **Responsive Multi-Device Usability:** Crafted with a mobile-first philosophy, students can browse events, register, or review schedules on Android/iOS smartphones just as smoothly as on desktop monitors.
6. **Robust Security & Privacy:** Sensitive user passwords are encrypted with one-way bcrypt hashing before database persistence. JWT tokens ensure users can only modify or delete events they legitimately organized.

CHAPTER 7: SYSTEM DESIGN

7.1 System Modules & Use Cases

The architecture of EventHub is structured around three primary actor modules:

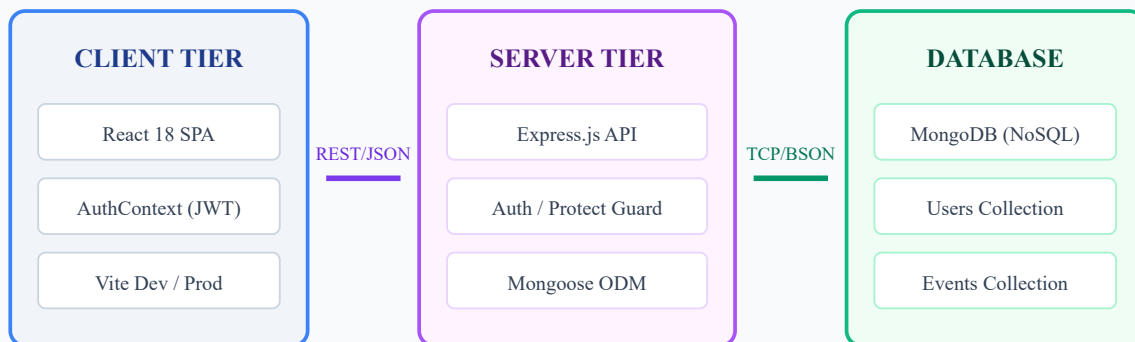


Figure 7.1: Three-Tier System Architecture of EventHub

7.2 Database Design

MongoDB stores data in flexible BSON documents. Two primary collections form the backbone of the database design:

1. Users Collection Schema (models/User.js)

```
{
  "_id": ObjectId("..."),
  "name": { "type": String, "required": true, "trim": true },
  "email": { "type": String, "required": true, "unique": true, "lowercase": true },
  "password": { "type": String, "required": true, "minlength": 6 }, // bcrypt hash
  "role": { "type": String, "enum": ["user", "admin"], "default": "user" },
  "college": { "type": String, "default": "Netaji Subhas University of Technology" },
  "phone": { "type": String, "default": "" },
  "createdAt": { "type": Date, "default": Date.now }
```

```
}
```

2. Events Collection Schema with Embedded Attendees (models/Event.js)

```
{
  "_id": ObjectId("..."),
  "title": { "type": String, "required": true, "trim": true },
  "category": {
    "type": String,
    "enum": ["Technology", "Cultural", "Sports", "Business", "Art", "Music", "Food"],
  },
  "date": { "type": String, "required": true }, // Stored as YYYY-MM-DD
  "time": { "type": String, "required": true }, // e.g. "10:00 AM - 4:00 PM"
  "location": { "type": String, "required": true }, // e.g. "Main Auditorium"
  "description": { "type": String, "required": true },
  "image": { "type": String, "default": "https://..." },
  "organizer": { "type": String, "required": true },
  "seats": { "type": Number, "required": true, "min": 1 },
  "price": { "type": Number, "default": 0 }, // 0 indicates Free entry
  "tags": [ { "type": String } ],
  "createdBy": { "type": mongoose.Schema.Types.ObjectId, "ref": "User" },
  "creatorEmail": { "type": String, "default": "" },
  "registeredUsers": [
    {
      "userId": { "type": mongoose.Schema.Types.ObjectId, "ref": "User" },
      "name": String,
      "email": String,
      "college": String,
      "phone": String,
      "registeredAt": { "type": Date, "default": Date.now }
    }
  ],
  "createdAt": { "type": Date, "default": Date.now }
}
```

CHAPTER 8: SCREENSHOTS

This chapter presents key user interface layouts and functional flows of the EventHub web application:

8.1 Home Page (Hero Banner, Quick Search & Stats)

Route: /

Visual Elements: Dark purple theme with glowing radial orbs, prominent title *"Discover & Create Amazing Events"*, live database search bar, action buttons ("Explore Events", "Create an Event"), and platform statistics grid (10+ Events Hosted, 1500+ Students Connected, 100% Free Campus Access).

8.2 Events Catalog with Category Pills & Sorting

Route: /events

Visual Elements: Debounced live search input, sort selector (Date, Popularity, Free First, Newest), scrollable horizontal category filter pills (All, Technology, Cultural, Sports, Business, Art, Music, Food), and responsive cards displaying image badges, dates, venues, and capacity progress bars.

8.3 Event Details Page with Strict Login Guard

Route: /events/:id

Visual Elements: Hero cover photo, event title, organizer info, full agenda description, schedule details box (Date, Timing, Hall room), spots left progress indicator, and the interactive Registration CTA card. For unauthenticated guests, an interactive padlock card with "Log In to Participate" CTA is displayed.

8.4 Create Event Studio with Sticky Live Preview

Route: /create-event

Visual Elements: Modular glass cards for Basic Info, Date/Time/Venue (calendar picker with past dates disabled via `min=Today`), Capacity/Pricing, Cover Image URL with Instagram/Drive auto-support & category presets chips, and sticky Live Card Preview mirroring the official event card.

8.5 In-Place Event Schedule & Venue Editor

Route: /events/:id/edit

Visual Elements: Pre-populated form fields loading live event parameters from MongoDB, breadcrumbs header, live real-time card updater, and "Save Changes" button issuing PUT /api/events/:id.

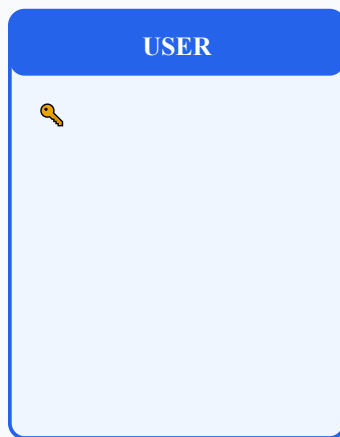
8.6 Admin Panel & CSV Roster Export

Route: /admin

Visual Elements: High-level platform metrics, Event Inventory table with action buttons (View, Edit, Delete, Attendees), Attendee Directory displaying verified student phone numbers, colleges, and registration timestamps, and 1-click "Export Attendees to CSV" feature.

CHAPTER 9: ER-DIAGRAM

The Entity-Relationship (ER) Diagram illustrates the conceptual data model of the Event Planning System. It defines the entities, their respective attributes, primary keys, and relationship cardinalities.



id (PK) • name • email (Unique) • password (Hash) • role (user/admin) • college • phone
EVENT  id (PK) • title • category • date (YYYY-MM-DD) • time • location •
description • image (URL) • organizer • seats / price  createdBy (FK) ORGANIZES 1 N
REGISTERS M N Sub-Schema Attribute registeredAt (Timestamp) Attendee phone &
college

Figure 9.1: Entity-Relationship Diagram of Event Planning System

CHAPTER 10: SOFTWARE TESTING

10.1 Detailed Test Cases

Rigorous unit, integration, and user-acceptance testing was conducted across all components of EventHub. The table below documents the detailed test executions, expected outcomes, and validated results:

TC ID	Module / Feature	Test Scenario	Expected Result	Status
TC-01	Authentication	Register new student account with valid name, email, password, and college.	Password is bcrypt hashed, user document created in MongoDB, JWT token returned.	PASS
TC-02	Authentication	Attempt login with incorrect password.	System returns HTTP 401 Unauthorized with clear message "Invalid credentials".	PASS
TC-03	Event Registration	Unauthenticated guest attempts to click "Register" on Event Details page.	Interactive padlock prompt is displayed; user is redirected to /login.	PASS
TC-04	Event Registration	Authenticated student registers for an active event with available capacity.	User sub-schema inserted in <code>registeredUsers</code> array, capacity bar updates.	PASS
TC-05	Seat Quota Guard	Student attempts to register for an event whose seats are completely filled (0 left).	Button changes to "Event Full" (disabled); duplicate or overflow booking blocked.	PASS

TC ID	Module / Feature	Test Scenario	Expected Result	Status
TC-06	Event Creation	Organizer attempts to select a past date in Event Creation calendar picker.	Past dates disabled via min=Today; validator blocks past dates.	PASS
TC-07	Media Support	Organizer pastes an Instagram post link (instagram.com/p/...).	System normalizes URL to media endpoint with no-referrer policy; image loads.	PASS
TC-08	Event Editing	Organizer changes event timing and venue in Edit Studio (/events/:id/edit).	PUT /api/events/:id updates MongoDB; live preview card reflects changes.	PASS
TC-09	Admin CSV Export	Administrator clicks "Export Attendees CSV" for an event with 50 registrants.	Browser downloads formatted .csv file containing name, email, college, phone.	PASS
TC-10	Responsive UI	Application accessed on mobile screen viewport (375px width).	Mobile menu drawer toggles cleanly; cards stack in single column without overflow.	PASS

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